

Full Stack Developer specializing in scalable system architecture and end-to-end design. Proven track record of delivering robust, user-centered solutions with a passion for writing clean and optimized code.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, Java, C, C#, SQL, Bash
- **Frontend:** React, Next.js, Angular, Tailwind CSS, HTML5, CSS3
- **Backend:** Node.js, Express.js, PostgreSQL, Valkey/Redis, MongoDB, REST APIs, WebSockets
- **Tools & DevOps:** Git, GitHub, Docker, AWS (App Runner, Secrets Manager), CI/CD, SSH, npm, Agile/Scrum
- **Concepts:** Object-Oriented Design, Asynchronous Programming, MVC Architecture, Responsive Design, User Research

WORK EXPERIENCE

Full Stack Developer

September 2025 - Present

Nuovo Artistic Photography; Montreal, QC

- Architect and maintain custom system integrations using AWS App Runner and Valkey to automate financial pipelines between Square, Xero, and other third-party APIs, improving data reliability as the company scales
- Identify and resolve workflow inefficiencies by designing and building internal tools, such as a skill-based task assignment system for retouchers and a Quality Assurance app for photographer training

Full Stack Developer

November 2023 - May 2025

Antimodular Research (Studio of Rafael Lozano-Hemmer); Montreal, QC

- Led development of 4+ interactive artworks exhibited globally and viewed by over 50,000 visitors, managing the full-stack architecture with minimal oversight and adapting to the artist's evolving vision
- Conducted research on human/object tracking, segmentation, and face recognition to improve new and existing projects

Research Intern

May 2021 - August 2021, September 2022 - December 2022

Autodesk; Toronto, ON

- Built full-stack and VR interface prototypes for internal UX research, such as a MERN scheduling app with real-time updates, still in use by the research team for coordinating weekly meetings

Teaching Assistant/Instructional Apprentice

January 2019 - August 2023

University of Waterloo (Sept 2021 - August 2023), Mount Allison University (Jan 2019 - May 2021)

- Led lab sessions and provided one-on-one support in office hours for introductory CS courses

EDUCATION

Master of Mathematics (MMath) in Computer Science

September 2021 - August 2023

University of Waterloo; Waterloo, ON

- Thesis-based master's program co-supervised by Prof. Daniel Vogel and Prof. Craig Kaplan (GPA = 95%)
- Thesis accepted for publication as a first-author paper at a top-tier computer science conference

Bachelor of Science (BSc), Computer Science & Mathematics, Physics Minor

September 2017 - May 2021

Mount Allison University; Sackville, NB

- Graduated with First Class Honours and distinction, Dean's List throughout (GPA = 91%)

AWARDS AND RECOGNITIONS

David R. Cheriton Graduate Scholarship (2021-2023): Awarded to top students based on academic excellence and research potential

L.A. Goodridge Scholarship (2017-2021): Awarded to one student per year level for exceptional academic standing

PUBLICATIONS

- **Saralin Zassman**, Craig S Kaplan, and Daniel Vogel. 2024. *Mindful Scroll: An Infinite Scroll Abstract Colouring App for Mindfulness*. Published at **CHI 2024**, the leading global conference in human-computer interaction, ACM.

LEADERSHIP EXPERIENCE

- **President**, Mathematics & Computer Science Society, Mount Allison University May 2020 - May 2021
- **Lead Coordinator**, Science Success Course, Mount Allison University September 2020 - May 2021

LANGUAGES

- Fluent in English and Mandarin; Conversational in French